Tecnologico de Monterrey

Interpartial Exam 1 B

2026-Jan-May

Grade

Course: Calculus I **ID-number:** ____________

Professor: Fco. Javier Vazquez Tavares **Date:** 26/01/2026

Last Names and Names: ______________________________ **Group:** 201

"*In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honestly and not to use unauthorized or illegal means to do so.*"

Signature: ____________

I Algebra Review. Read the following questions and justify your answer with a procedure.

1. (10 **points**) Solve the following equation,

$$f(x) = e^{2x} - 3e^{x} = -2$$

2. (12 **points**) Find the asymptotes and intersections of the following function,

$$g(x) = \frac{4x^2 - 1}{2x^2 - 1}$$

3. (12 **points**) Consider that $h(x) = e^x$ and $j(x) = 1 - x$. Find the following operations,

(a) $m(x) = (h \circ j)(x)$ (b) $n(x) = (j \circ h)(x)$ (c) Find the inverse funciton of $m(x)$

$m(x)$

$n(x)$

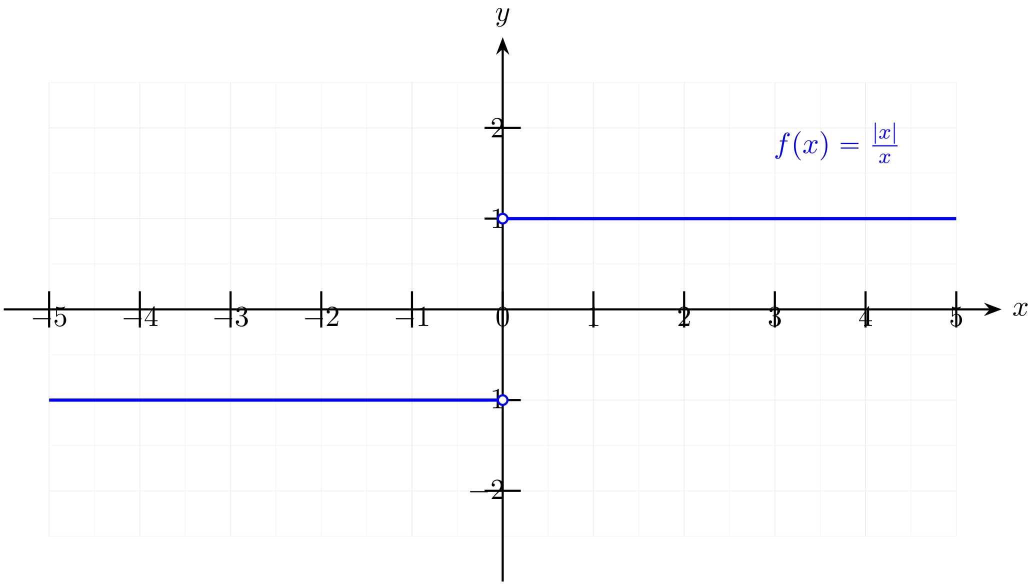
$m^{-1}(x)$

II Limits. Read the following questions and justify your answer with a procedure.

1. (5 **points**) Identify the discontinuities of the function (vertical asymptotes or holes) and compute the limit using the numeric approach at those values of the function $f(x) = \frac{x^2-4}{x-2}$.

2. (5 **points**) Identify the discontinuities of the function (vertical asymptotes or holes) and compute the limit using the numeric approach at those values of the function $f(x) = \frac{x^2-4}{x-2}$.

3. (5 **points**) Applying the graphical technique to compute limits, argument if the limit converges of the following functions,



4. (5 **points**) Applying the graphical technique to compute limits, argument if the limit converges of the following functions,

1. (5 **points**) ______ Select the option with a quadratic function:

   (a) $(a/b)x + c$ (b) $(a_1/b_1)x + (a_2/b_2)x^2$ (c) $(a/b)x^3 + c$ (d) $(a_n/b_n)x^{1/2}$

2. (5 **points**) ______ Select the correct domain, such that $f(x) \in \mathbb{R}$, where $f(x) := x^{1/n}$

   (a) $x \in (-\infty, \infty)$ (b) $x \in [0, \infty)$ (c) $x \in (-\infty, 0]$ (d) $x \in [-1, 1]$

3. (5 **points**) ______ Select the option with the range of the function $f(x) := \mathrm{abs}(x)$ for $x \in (-\infty, \infty)$

   (a) $f(x) \in (-\infty, \infty)$ (b) $f(x) \in [0, \infty)$ (c) $f(x) \in (-\infty, 0]$ (d) $f(x) \in [-1, 1]$

4. (5 **points**) ______ Select the correct type of function of $f(x) := x^{1/n}$, where $n \in \mathbb{Z}$ (x belongs to the set of integer numbers)

   (a) Quadratic (b) Cubic (c) Exponential (d) Rational

5. (5 **points**) ______ Select the alternative mathematical notation to express $g[f(x)]$

   (a) $f \circ g$ (b) $g \circ f$ (c) $f \to g$ (d) $f(g)$

6. (5 **points**) ______ Select the option of the function that map all the domain into the positive real numbers.

   (a) $f(x) := x^3$ (b) $f(x) := \mathrm{abs}(x)x$ (c) $f(x) := \mathrm{sgn}(x)x$ (d) $f(x) := -x^2$

II Solve the following exercises in an orderly and clear manner. **Answers without procedure will not be considered.** Underline or frame your final answer.

1. (12 **points**) Tacking into account the following functions,

$$f(x) := \frac{a}{b}x + c, \quad g(x) := x + h,$$

express the algebraic representation of

$$H(x) := \frac{(f \circ g)(x) - f(x)}{h}.$$

2. (12 **points**) Tacking into account the following functions,

$$f(x) := \frac{a}{b}x, \quad g(x) := \frac{c}{d}x^2,$$

represent the algebraic expression of,

$$\frac{f(x)}{g(x)}, \quad \frac{g(x)}{f(x)}.$$

Finally, write the main difference between them.

3. (12 **points**) Use the verbal representation of the following functions,

(a) $f(x) := ax + c$ (b) $f(x) := \left(a^2 + b^2\right)^{1/2}$ (c) $f(x) := (x-1)^{1/2}, x \in (1, 10]$

4. (12 **points**) Write a general algebraic expression for each graph,

5. (12 **points**) Graph a sketch of the following function,

$$f(x) := \begin{cases} x^2 & x \in [0, \infty) \\ -x^2 & x \in (-\infty, 0] \end{cases}$$

6. (10 **points**) Express the domain and range of the following functions,

(a) $f(x) := (x+1)^{1/2}$ (b) $g(x) := 1/x^2$